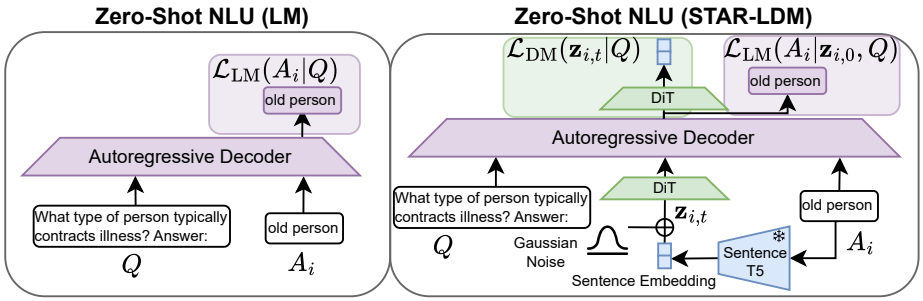

Figure 3: Diagram comparing zero-shot NLU evaluation for standard LMs (left) and STAR-LDM (right). Both models score candidate answers (A_i) given a question (Q). STAR-LDM utilizes both the answer text A_i and its latent embedding $\mathbf{z}_{i,0}$ from Sentence-T5 for scoring.

5 Natural Language Understanding

Standard Natural Language Understanding (NLU) benchmarks, particularly multiple-choice question answering (QA) datasets (e.g., CSQA, SIQA, ARC), assess a model's ability to identify the most plausible answer or completion (A) from a set of candidates, given a context or question (Q). Conventional autoregressive models (Figure 3, Left Panel) typically score each candidate answer A_i by its negative log-likelihood: $\mathcal{L}_{\text{LM}}(A_i|Q) = -\log p(A_i|Q)$. This likelihood is calculated using teacher forcing, and the candidate with the lowest score (highest probability) is selected as the answer.

We adapt this protocol for STAR-LDM to select the most likely candidate answer given the question Q (Figure 3, Right Panel). We assess candidate answers by approximating $-\log p(A_i|Q)$ using the variational lower bound on the likelihood (Kingma et al., 2021), as direct calculation is intractable. We encode each candidate A_i using Sentence T5 to produce a semantic latent plan $\mathbf{z}_{i,0}$, add Gaussian noise to generate $\mathbf{z}_{i,t}$ at a specific timestep t, and compute our scoring function derived from the Evidence Lower Bound (ELBO):

$$
\begin{aligned}
-\log p(A_i|Q) &\leq \mathcal{L}_{\text{LM}}(A_i|\mathbf{z}_{i,0}, Q) + \mathcal{L}_{\text{DM}}(\mathbf{z}_{i,t}|Q) \qquad (10)\\
&= -\sum_{j=1}^{|A_i|} \log p(A_i^j|A_i^{<j}, \mathbf{z}_{i,0}, Q) + \mathbb{E}_{t,\epsilon\sim\mathcal{N}(0,I)}[w_{\text{ELBO}}(\lambda_t) \cdot ||\epsilon - \epsilon_\theta(\mathbf{z}_{i,t}, t, Q)||_2^2]
\end{aligned}
$$

where the weighting $w_{\text{ELBO}}(\lambda_t)$ for the regression objective follows from the derivation of the lower bound (Kingma et al., 2021). We present a detailed derivation of this scoring function in Appendix C. The LM term measures how well the model reconstructs A_i given Q and $\mathbf{z}_{i,0}$, while the diffusion term measures how plausible the latent plan $\mathbf{z}_{i,0}$ is for question Q. We select the candidate with the minimum overall score, choosing the answer that is both linguistically well-formed and has a plausible semantic meaning. We compute the expectation in Equation 10 with an unbiased Monte Carlo estimate of the weighted reconstruction error. In practice, autoregressive models often normalize the log-likelihood, $\mathcal{L}_{\text{LM}}$, by the length of the candidate A_i to improve performance. We also adopt this approach as we observe it similarly improves performance.

Results. STAR-LDM outperforms similarly sized autoregressive models on standard zero-shot NLU benchmarks[2], as shown in Table 1. The model excels particularly on commonsense reasoning tasks (CSQA, SIQA), demonstrating the effectiveness of latent semantic planning. Notably, the diffusion component alone performs strongly, indicating that semantic plausibility contributes significantly to reasoning capabilities. While the autoregressive component underperforms independently, combining it with diffusion yields the best results, suggesting complementary strengths. This demonstrates that our diffusion planning enhances the NLU capabilities of comparable autoregressive models.

[2]The benchmarks include CommonsenseQA (CSQA), Social IQA (SIQA), HellaSwag (HS), Winogrande (WG), Physical IQA (PIQA), OpenBookQA (OBQA), and ARC (Easy and Challenge).

Model	Params	Scoring	CSQA	SIQA	HS	WG	PIQA	OBQA	ARC:E	ARC:C	Avg
Random Baseline	—		20.0	33.3	25.0	50.0	50.0	25.0	25.0	25.0	31.7
GPT2-Large	.77B	LM Loss	36.6	42.1	42.9	51.9	69.2	34.2	46.6	25.1	43.6
GPT2-XL	1.5B	LM Loss	36.3	42.3	47.9	53.1	70.5	34.4	51.1	28.5	45.5
Pytha-1b	1B	LM Loss	35.3	42.4	45.6	52.6	69.2	32.4	49.1	27.0	44.2
Pytha-1.4b	1.4B	LM Loss	35.7	43.2	50.9	54.2	71.1	36.0	54.0	28.3	46.7
STAR-LDM	.96B	LM Loss	$35.2_{0.4}$	$42.2_{0.2}$	$44.0_{0.1}$	$50.5_{0.1}$	$69.3_{0.1}$	$27.1_{0.3}$	$37.6_{0.5}$	$26.3_{0.3}$	$41.5_{0.1}$
		Diffusion Loss	$48.7_{0.3}$	$46.8_{0.5}$	$37.9_{0.1}$	$50.4_{0.2}$	$65.5_{0.2}$	$39.3_{0.1}$	$55.0_{0.6}$	$32.1_{0.5}$	$47.0_{0.1}$
		LM + Diff Loss	$49.8_{0.3}$	$46.8_{0.3}$	$39.0_{0.1}$	$51.4_{0.1}$	$66.1_{0.1}$	$41.3_{0.2}$	$55.0_{0.3}$	$31.4_{0.8}$	$47.6_{0.1}$

Table 1: Zero Shot NLU Evaluation Results. We report mean and standard error over three monte-carlo evaluations for STAR-LDM.

6 StoryCloze Generation

Experimental Setup. To evaluate generation quality, we assess narrative continuations on the StoryCloze dataset (Mostafazadeh et al., 2016), which requires coherent fifth-sentence completions for four-sentence stories. We employ Claude 3.7 Sonnet (Anthropic, 2025) as a judge to compare STAR-LDM against baseline models (GPT-2 Large/XL (Radford et al., 2019) and Pythia models (Biderman et al., 2023)) in blind head-to-head evaluations across 200 contexts, using four criteria: Narrative Coherence, Commonsense Reasoning, Language Quality, and Emotional/Psychological Plausibility. Full evaluation details, including the prompt, are in Appendix D.

Results. STAR-LDM significantly outperforms similar-sized and larger models across all criteria (Figure 4), with win rates exceeding 70% against models like GPT-2 XL on Coherence/Reasoning ($p < 0.01$). These substantial gains suggest our latent diffusion planning effectively improves global structure and logical consistency compared to pure autoregression. While competitive against much larger models (Pythia 6.9/12B), STAR-LDM's improvements in Language Quality are less pronounced, indicating the planning mechanism primarily enhances semantic coherence rather than surface fluency.

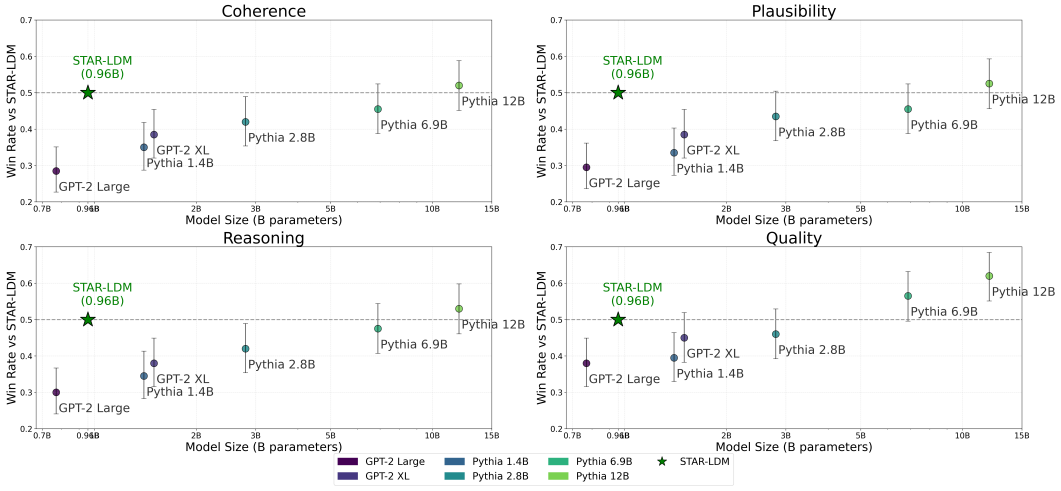

Figure 4: LLM-Judge evaluation. Results presented with 95% confidence intervals.

7 Language Generation Evaluation

Experimental Setup. We evaluate generation on 5000 C4 validation samples (Raffel et al., 2019), generating 64-token continuations from 32-token prefixes. We evaluate using three complementary metrics: (1) **Llama Ppl** measures fluency using perplexity computed by Llama-3.2-3B (Grattafiori et al., 2024); (2) **MAUVE** (Pillutla et al., 2021) assesses distributional similarity to human-written text using GPT-2 Large embeddings; and (3) **Div** quantifies

			C4 Validation		
	Params	CFG (w)	MAUVE ↑	Generative Perplexity ↓	Diversity ↑
GPT-2 Large	.77B	-	$85.2_{0.6}$	$30.5_{0.3}$	$41.8_{0.1}$
GPT-2 XL	1.5B	-	$86.6_{0.4}$	$27.1_{0.6}$	$43.3_{0.1}$
Pythia-1b	1B	-	$85.5_{0.1}$	$36.5_{0.7}$	$45.6_{0.1}$
Pythia 1.4b	1.4B	-	$84.8_{0.5}$	$33.1_{.1}$	$47.1_{0.0}$
STAR-LDM	.96B	1.0	$94.6_{0.2}$	$36.1_{0.5}$	$45.3_{0.0}$
STAR-LDM	.96B	1.5	$94.1_{0.2}$	$31.1_{0.5}$	$45.4_{0.0}$
STAR-LDM	.96B	2.0	$94.0_{0.2}$	$29.8_{0.6}$	$45.4_{0.0}$

Table 2: Language generation evaluation on C4 validation examples. We report mean and standard error over three sets of generations.

lexical diversity as $\prod_{n=2}^{4} \frac{|\text{unique n-grams}|}{|\text{total n-grams}|}$ (Su et al., 2022). For STAR-LDM, we report metrics across different classifier-free guidance strengths (w), which increase the influence of the prefix, during diffusion planning.

Results. As shown in Table 2, STAR-LDM consistently outperforms baselines in MAUVE score, while maintaining competitive perplexity and diversity. Increasing prefix guidance improves fluency without sacrificing diversity. We present results regarding inference latency in Appendix B.

8 Plug-and-Play Control

Experimental Setup. STAR-LDM enables attribute steering during generation without model retraining. We evaluate sentiment control and toxicity mitigation and compare against DeXperts (Liu et al., 2021a)—a strong baseline that fine-tunes specialized language models to guide generation. In contrast, our approach only requires training a lightweight MLP classifier on the target attribute to guide diffusion planning.

For toxicity control, we trained our MLP on the Jigsaw Unintended Bias dataset (cjadams et al., 2019). The toxicity guidance was evaluated using 1,000 randomly selected neutral prompts from RealToxicityPrompts (Gehman et al., 2020). For sentiment control, we used both Amazon Polarity [3] and Stanford Sentiment Treebank (SST-2) (Socher et al., 2013) to train our sentiment MLP classifier. We tested sentiment steering using 1,000 randomly selected neutral prompts from OpenWebText (Gokaslan et al., 2019; Liu et al., 2021a).

For evaluation in both settings, we generated 10 continuations per prompt. We measure the alignment with the control condition, language fluency, and language diversity. For toxicity, we follow prior work Deng & Raffel (2023); Liu et al. (2021a)and use the Perspective API. For sentiment we utilize RoBERTa-Large[4] (Liu et al., 2020) fine-tuned on sentiment classification across diverse domains. For language fluency, we report the generative perplexity with Llama-3.2-3B. To quantify generation diversity, we follow prior work (Lu et al., 2022b) and measure the average number of unique 3-grams in each set of continuations.

Results. Figure 5 shows STAR-LDM achieves better control-fluency trade-offs than DeXperts across guidance scales for both negative sentiment and toxicity reduction. Our model generates near-zero positive sentiment text at lower perplexity than GPT-2 Large, while maintaining lexical diversity even under strong guidance (additional results presented in Appendix E).

[3] https://huggingface.co/datasets/amazon_polarity

[4] https://huggingface.co/siebert/sentiment-roberta-large-english